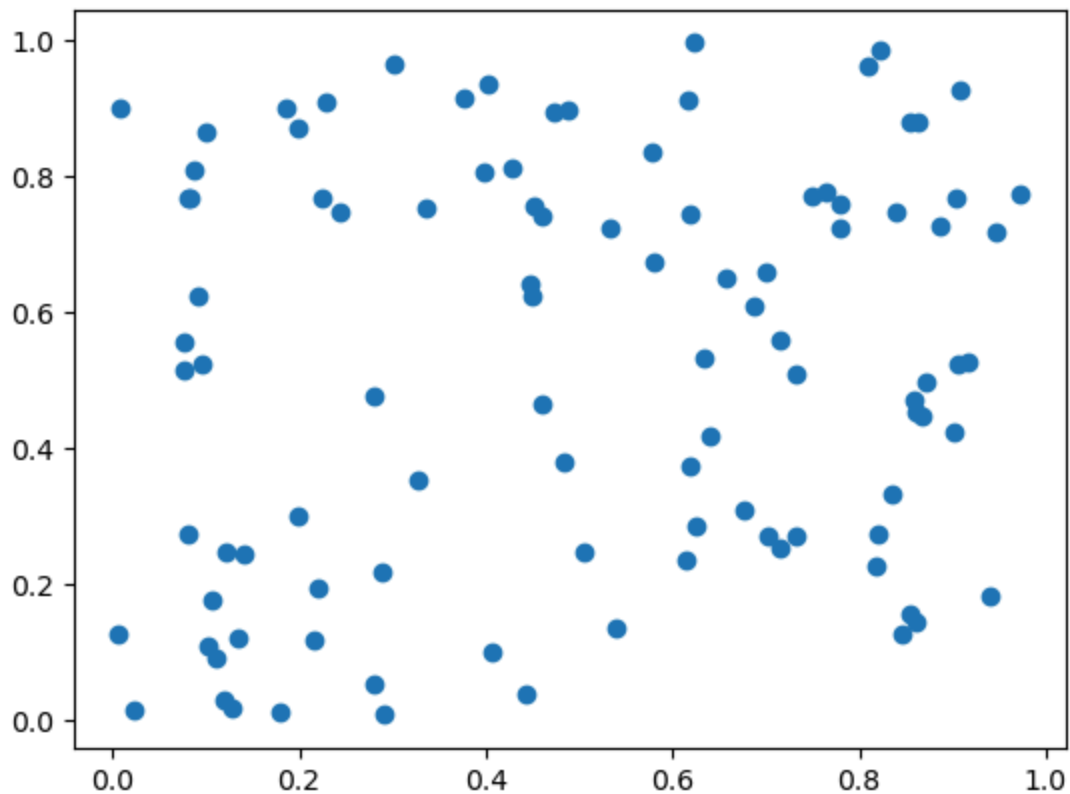


```
In [1]: import numpy as np
        from sklearn.cluster import KMeans
        import matplotlib.pyplot as plt
```

```
In [2]: X=np.random.rand(100,2)
```

```
In [3]: plt.scatter(X[:,0], X[:,1])
        plt.show()
```



```
In [4]: kmeans=KMeans(n_clusters=3, random_state=0)
        kmeans.fit(X)
```

D:\gta\anaconda\Lib\site-packages\sklearn\cluster_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.

warnings.warn(

```
Out[4]:
```

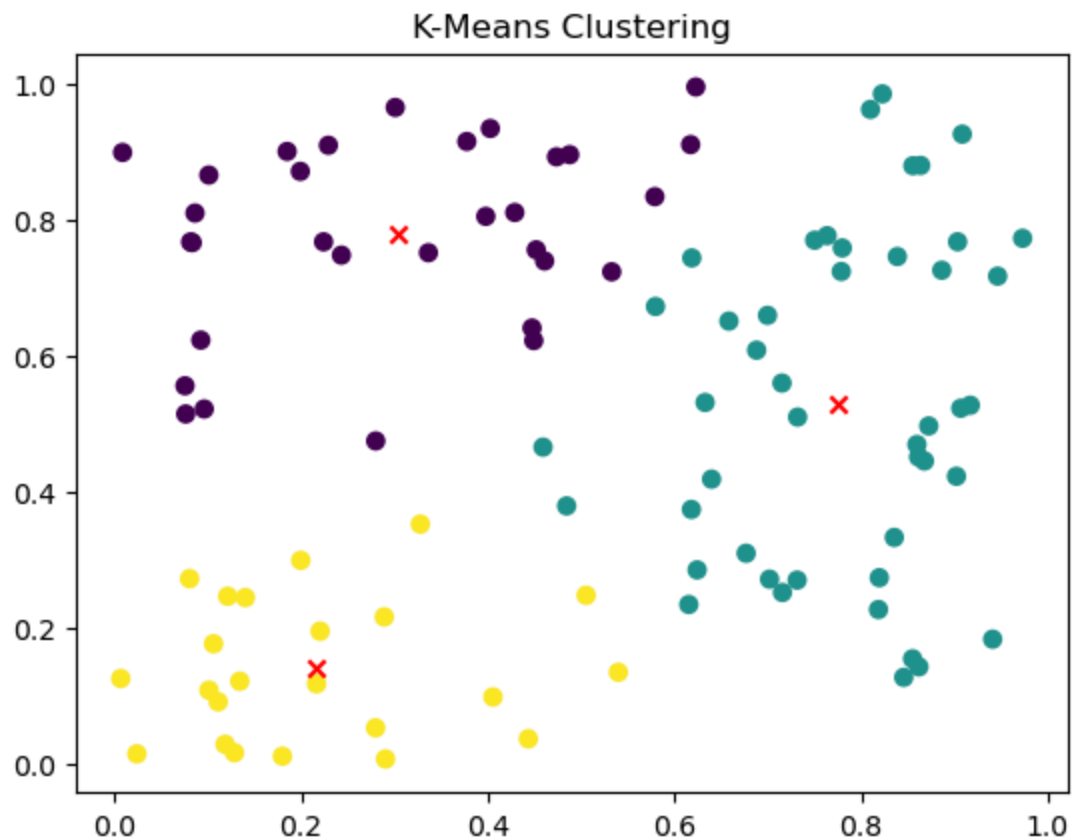
▼ KMeans ⓘ ?

► Parameters

```
In [5]: centroids=kmeans.cluster_centers_
        labels = kmeans.labels_
```

```
In [6]: plt.scatter(X[:, 0], X[:,1], c=labels)
        plt.scatter(centroids[:, 0], centroids[:, 1], marker='x', color='red')
```

```
plt.title("K-Means Clustering")  
plt.show()
```



In []:

In []: